



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 6 Name: Dhanish Ladwani PRN: 24070521020

1. Experiment Title:

Use string functions and regex for email validation, data extraction, and text analysis.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Experiment 6.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>String Methods & Regular Expressions</title>
  <link rel="stylesheet" href="Experiment 6.css">
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
</head>
<body>
  <div class="container">
    <h1>String Methods & Regular Expressions</h1>
    <div class="box">
      <div class="input-group">
        <label for="paragraph"><b>Enter a paragraph:</b></label>
        <textarea id="paragraph" rows="5">Javascript is a powerful programming language. It
is widely used for web development.</textarea>
      </div>

      <div class="input-group">
        <label for="email"><b>Enter Email:</b></label>
        <input type="text" id="email" placeholder="Enter your email">
      </div>

      <button onclick="processString()">Process</button>
      <div id="output"></div>
    </div>

    <footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-
top: 1px solid lightgray; padding-top: 15px;">
      Made by Dhanish Ladwani PRN - 24070521020
    </footer>
  </div>

  <script src="Experiment 6.js"></script>
</body>
</html>
```

Experiment 6.css

```
:root {
  --primary: #2563eb;
  --primary-hover: #1d4ed8;
  --bg-main: #f8fafc;
  --surface: #ffffff;
  --text-primary: #1e293b;
  --text-secondary: #64748b;
  --border-color: #e2e8f0;
  --radius: 10px;
  --shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.05), 0 2px 4px -2px rgba(0, 0, 0, 0.05);
}

* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  background-color: var(--bg-main);
  color: var(--text-primary);
  font-family: 'Inter', system-ui, -apple-system, sans-serif;
  line-height: 1.5;
  padding: 40px 20px;
  display: flex;
  justify-content: center;
}

.container {
  width: 100%;
  max-width: 800px;
  background-color: var(--surface);
  padding: 32px;
  border: 1px solid var(--border-color);
  border-radius: 16px;
  box-shadow: var(--shadow);
}

.app-header {
  text-align: center;
  margin-bottom: 28px;
}

.app-header h1 {
  color: #1e3a8a;
  font-size: 26px;
  font-weight: 700;
  margin-bottom: 6px;
}

.subtitle {
  color: var(--text-secondary);
  font-size: 14px;
}

.box {
  display: flex;
  flex-direction: column;
  gap: 18px;
}

.input-group {
  display: flex;
  flex-direction: column;
  gap: 6px;
}

label {
```

```

    font-size: 14px;
    color: var(--text-primary);
}

input, textarea {
    padding: 12px;
    width: 100%;
    border: 1px solid var(--border-color);
    border-radius: var(--radius);
    font-family: inherit;
    font-size: 14px;
    outline: none;
    transition: all 0.2s ease;
    background-color: #f8fafc;
}

input:focus, textarea:focus {
    border-color: var(--primary);
    background-color: #ffffff;
    box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.15);
}

.action-group {
    margin-top: 4px;
}

button {
    padding: 12px 24px;
    background-color: var(--primary);
    color: white;
    border: none;
    border-radius: var(--radius);
    font-size: 15px;
    font-weight: 600;
    cursor: pointer;
    transition: background-color 0.2s ease, transform 0.1s ease;
}

button:hover {
    background-color: var(--primary-hover);
}

button:active {
    transform: scale(0.98);
}

.output-container {
    margin-top: 10px;
}

.results-grid {
    display: flex;
    flex-direction: column;
    gap: 16px;
}

.card {
    background: #f8fafc;
    border: 1px solid var(--border-color);
    border-left: 4px solid var(--primary);
    border-radius: var(--radius);
    padding: 16px;
    transition: transform 0.15s ease, box-shadow 0.15s ease;
}

.card:hover {
    transform: translateY(-2px);
    box-shadow: var(--shadow);
}

```

```
.card h3 {
  color: #1e3a8a;
  font-size: 15px;
  font-weight: 600;
  margin-bottom: 8px;
}

.card p {
  font-size: 14px;
  color: var(--text-primary);
  word-break: break-word;
}

.card small {
  display: block;
  margin-top: 6px;
  color: var(--text-secondary);
  font-size: 12px;
}

.code-block {
  font-family: monospace;
  background: #e2e8f0;
  padding: 6px 10px;
  border-radius: 6px;
  font-size: 13px;
  color: #0f172a;
}

.reversed-text {
  font-family: monospace;
  color: #475569;
  letter-spacing: 1px;
}

.badge {
  display: inline-block;
  padding: 4px 10px;
  border-radius: 20px;
  font-size: 12px;
  font-weight: 600;
}

.badge.success {
  background-color: #dcfce7;
  color: #166534;
}

.badge.danger {
  background-color: #fee2e2;
  color: #991b1b;
}

.error-card {
  border-left-color: #ef4444;
  background-color: #fef2f2;
}

.error-msg {
  color: #b91c1c;
  font-weight: 500;
}

.app-footer {
  text-align: center;
  margin-top: 32px;
  font-size: 13px;
  color: var(--text-secondary);
}
```

```
border-top: 1px solid var(--border-color);
padding-top: 16px;
}
```

Experiment 6.js

```
function processString() {
    let paragraph = document.getElementById("paragraph").value;
    let email = document.getElementById("email").value;

    let words = paragraph.split(/\s+/);

    let vowels = paragraph.match(/[aeiou]/gi);
    let vowelCount = vowels ? vowels.length : 0;

    let replacedParagraph = paragraph.replace(
        /JavaScript/gi,
        "JavaScript Programming"
    );

    let searchWord = "powerful";
    let position = paragraph.indexOf(searchWord);

    let emailRegex =
        /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;

    let emailResult;

    if(emailRegex.test(email)) {
        emailResult = "Valid Email Address";
    } else {
        emailResult = "Invalid Email Address";
    }
    let emailText =
        "For queries contact student@example.com or admin@college.edu";

    let extractedEmails = emailText.match(
        /[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}/g
    );

    let reeversedParagraph = paragraph.split("").reverse().join("");

    document.getElementById("output").innerHTML = `
    <h3>1. Original Paragraph</h3>
    <p>${paragraph}</p>

    <h3> 2. split() - Words </h3>
    <p> ${words.join(", ")}</p>

    <h3> 3. match() - Vowels </h3>
    <p> ${vowels ? vowels.join(", ") : "NO VOWELS FOUND!"}</p>

    <h3> 4. Vowel Count </h3>
    <p> Total number of vowels: <b>${vowelCount}</b></p>

    <h3> 5. replace() - Replace Text </h3>
    <p> ${replacedParagraph} </p>

    <h3> 6. indexOf() - Search Word </h3>
    <p>
        Position of "<b>${searchWord}</b>":
        <b>${position}</b>
    </p>

    <h3> 7. Email Validation using Regex </h3>
    <p>
```

```

    Email: <b>${email}</b><br>
    Result: <b>${emailResult}</b>
  </p>

  <h3> 8. Regex - Extracted Information </h3>
  <p>
    ${extractedEmails ? extractedEmails.join("<br>") : "NO EMAIL ADDRESS FOUND!"}
  </p>

  <h3> 9. Reversed Paragraph </h3>
  <p> ${reeversedParagraph} </p>
  `;
}

```

4. Output:

The screenshot displays a web application titled "String Methods & Regular Expressions". It features two input fields: "Enter a paragraph:" with the text "Javascript is a powerful programming language. It is widely used for web development." and "Enter Email:" with the text "dhanishladwani@gmail.com". A blue "Process" button is located below the email input.

The output section lists nine string methods and their results:

- 1. Original Paragraph**: Javascript is a powerful programming language. It is widely used for web development.
- 2. split() - Words**: Javascript,is,a,powerful,programming,language.,It,is,widely,used,for,web,development.
- 3. match() - Vowels**: a, a, i, i, a, o, e, u, o, a, i, a, u, a, e, l, i, i, e, u, e, o, e, e, e, o, e
- 4. Vowel Count**: Total number of vowels: 27
- 5. replace() - Replace Text**: JavaScript Programming is a powerful programming language. It is widely used for web development.
- 6. indexOf() - Search Word**: Position of "powerful": 16
- 7. Email Validation using Regex**: Email: dhanishladwani@gmail.com
Result: Valid Email Address
- 8. Regex - Extracted Information**: student@example.com
admin@college.edu
- 9. Reversed Paragraph**: .tnempoleved bew rof desu ylediW si tI .egaugnaI gnimmargorp lufrewop a si tpirsavaJ

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5. Result/Conclusion:

A web application was built to demonstrate JavaScript string methods and regular expressions. The `split()` method was used to break a paragraph into words, `match()` was used with a regular expression to extract vowels and count them, and `replace()` was used to substitute a keyword throughout the text. The `indexOf()` method located the position of a specific word, while regular expressions validated an entered email address and extracted all valid email addresses from a block of text. The split-reverse-join technique was also used to reverse the paragraph. This experiment reinforced string manipulation and pattern matching using regular expressions in JavaScript.